Fahim Al Islam

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Education

B.Sc. in Computer Science and Engineering *CGPA: 3.4 out of 4*
University of Rajshahi, Rajshahi

Work Experience

iOS Developer *Jan 2026 – Present*
MyMedicalHub International

- Architected a real-time pose estimation iOS application using **SwiftUI**, **MVVM**, **TensorFlow Lite (MoveNet)**, and **AVFoundation**, processing live camera frames at **30 FPS**. Built a multi-tenant assessment platform powering App Store products including **EMMA Health**, **Atlas Athlete**, and **AI Guided Therapy**, featuring configurable exercise workflows, audio-guided assessments, and state-machine-driven movement evaluation.
- Implemented pose validation and user orientation detection using body landmark geometry to ensure assessment quality and prevent invalid recordings. Developed generation-based timer protection to safely manage asynchronous countdown and recording workflows.
- Optimized the camera pipeline by suspending ML inference during recording, reducing CPU/GPU contention. Built non-blocking audio coordination, body-visibility buffering, and event rate-limiting mechanisms to improve tracking stability and overall user experience.

Personal Projects

CreatorStudioPro iOS App(*SwiftUI,UiKit,MVVM,AVFoundation*)

- Built a modular **iOS media engine** in **Swift 6** using **AVFoundation**, **Vision**, **Speech**, and **CoreML**. Implemented a **UIKit–SwiftUI bridge**, coordinator-based navigation, real-time face analysis (**10fps**), OCR-based expiry date detection, and barcode scanning. Solved complex camera, `XPC`, and `Combine` issues while improving app reliability, performance, and maintainability. (GitHub) (Demo)

AgriBangla iOS App(*SwiftUI,UiKit,MVVM,Coordinator Pattern,Firebase,Cloudinary*)

- Developed a role-based iOS marketplace app using **SwiftUI**, **MVVM**, **Coordinator**, and **Firebase Firestore**, featuring product listings, cart checkout, order management, **Google Sign-In**, and email authentication. Implemented secure atomic order processing with Firestore batch writes, balance/stock updates, and role-based access control.
- Built an on-device plant disease detection app using custom **TensorFlow Lite** model with real-time camera inference, AI-powered report generation (**Gemini** and **DeepSeek**), Delta-T spray recommendations, and PDF export. Improved reliability through dependency injection, API fallback strategies, and Firestore data consistency fixes. (GitHub) (Demo)

Programming Achievements

- **58th** out of 1700+ teams in the **ICPC** preliminary contest and participated onsite contest (RU_Uninspired)
- **48th** out of 1000+ teams in the **NCPC** preliminary contest and participated in the onsite contest. (RU_Riddle)
- **61st** out of 150+ teams in the KUET IUPC onsite contest (RU_Igniters)

Online Programming Competitions

- **Codeforces:** _Fahim9090 Max Rating: **Expert (1665)**
- **LeetCode:** _Fahim9090 Max Rating: **Knight (1897)** *(Top 6% Worldwide)*
- **Codechef:** _fahim124 Max Rating: **4* (1863)**
- **Other Judge:** *Solved **2500+** problems in different online judges*

Technical Skills

- **Programming Languages:** C/C++, Python, Java, **Swift**, **Objective-C**
- **Frameworks, Tools & Concepts: SwiftUI**, **UiKit**,Flutter,Android Studio,Git, MySQL